



Review of the lethal mechanism of insulin poisoning and the characteristic of forensic identification

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ABSTRACT

Insulin, as the only hypoglycemic hormone in the body, plays a key role in blood sugar control. However, excessive insulin intake can lead to insulin poisoning and even death, which often occurs in clinical and forensic work. At present, some researches on insulin poisoning have been carried out at home and abroad, however, it seems that the mechanism and forensic characteristics of insulin poisoning are not clear and complete. Therefore, in this paper, we reviewed the potential mechanism of insulin poisoning, the methods of insulin detection and the forensic identification of poisoning cases, aiming at providing services for the forensic identification of insulin poisoning.

1. Insulin and insulin analogues

Human insulin is the only hypoglycemic hormone synthesized by human islet beta cells with a relative molecular weight of 5807 Da. It is formed by a 21-amino acid peptide chain A and a 30-amino acid peptide chain B connected by two interchain disulfide bonds (A7-B7, A20-B19) [1]. In 1921, dog insulin was first discovered to have a hypoglycemic effect. However, due to the low extraction efficiency of animal insulin and easy immune reaction, a variety of insulin analogues (hereinafter collectively referred to as insulin) produced by engineering technology are gradually applied in the medical field [2], for example, insulin lispro was produced after the exchange of position 28 proline and position 29 lysine of human insulin B chain, and insulin aspart was produced after replacing position 28 proline of B chain with aspartic acid. Nowadays, insulin is usually divided into four categories according to the duration of its action: rapid-acting, short-acting, medium-acting and long-acting insulin (Table 1), and the action time varies from a few hours to tens of hours [3]. For a large number of insulin types, patients should comprehensively consider their own blood sugar, whether there are complications, and the doctor's treatment plan and other factors, choose their own insulin preparations to prevent the occurrence of insulin poisoning.

2. Potential mechanisms of insulin poisoning

The fasting blood glucose content of healthy people is 3.9 ~ 6.1 mmol/L. After diet, the body's blood glucose concentration and insulin secretion increase, and the metabolism of sugar, fat and protein is adjusted to maintain normal blood glucose concentration, and then metabolism through the liver and kidney is inactivated [4,5]. When the body injected a large amount of insulin in a short time, exceeding the metabolic capacity of the liver and kidney, the blood sugar concentration of the body dropped significantly, which may cause dizziness, palpitations, confusion and other symptoms, and even coma or death in severe cases [6,7]. At present, the mechanism of insulin poisoning has not been pointed out fully and clearly, and some existing studies [8–13] suggested that it may be involved in the central nervous system and the cardiac conduction system, which was shown in Fig. 1.

2.1. Hypoglycemic brain injury

Hypoglycemic brain damage is thought to be one of the causes of death from insulin poisoning [14,15]. Early studies generally believed that insulin induced hypoglycemia brain injury was caused by excitatory neurotoxicity. In rat model of insulin-induced severe hypoglycemia, the rats mostly showed convulsions and seizures, and the

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electroencephalogram showed abnormal discharge in multiple regions [16]. In addition, people often showed mental and emotional changes under hypoglycemia [17], suggesting that hypoglycemia may lead to changes in some neurotransmitters in the brain. In this regard, some scholars found that glutamate content in the brain increased significantly after glucose deprivation [18], and PARAMO et al [19] built a hypoglycemic brain injury cell model using cultured hippocampal neurons in vitro, and also found that the extracellular glutamate level of neurons was significantly upregulated, and glutamate, as an important excitatory neurotransmitter in the brain, was significantly upregulated. The significant change of its content indicated the obvious correlation between excitatory neurotoxicity and hypoglycemic brain injury.

In addition, in the process of the gradual lack of glucose in the brain, the entry of aerobic metabolites into the tricarboxylic acid cycle is reduced, and oxaloacetic acid and glutamic acid accumulated in mitochondria generate a large amount of aspartic acid under the action of aminotransferase. Methyl-d-aspartic acid, glutamate receptor, N-methyl-D-aspartic acid (NMDA), the combination of NMDA and α -amino-3-hydroxy-5-methyl-4-isoxazole-propionic acid (AMPA) increases excitatory neurotoxicity [19]. After intraventricular injection of the N-methyl-D-aspartic acid receptor antagonist dizocilpine, it was found that the neuronal necrosis of insulin-induced hypoglycemic brain injury was significantly reduced, indicating the important role of excitatory neurotoxicity in insulin-induced hypoglycemic brain injury.

Except for the excitatory neurotoxicity, some scholars believe that oxidative stress injury is also involved in the occurrence and development of hypoglycemic brain injury. It was found that hypoglycemic brain injury mainly caused the necrosis of superficial cortical neurons (layers 2 and 3), especially the degeneration and necrosis of granulos cells in hippocampal dentate gyrus (DG) [20–22]. In addition, the content of lipid peroxides and markers of oxidative stress in the cerebral cortex and hippocampus were significantly increased, and the up-regulation of glutamate was found to be closely related to the increase of oxidative stress products, which may be one of the main reasons for the necrosis of DG in hippocampus after hypoglycemic brain injury [23]. Other studies have found that glutamate binds to NMDA and/or AMPA receptors to induce a large number of calcium ions (Ca^{2+}) to flow in the environment of glucose deprivation, leading to mitochondrial membrane depolarization and Ca^{2+} overload, and at the same time activating reduced Coenzyme II oxidase and xanthine oxidase in neurons. Reactive oxygen species (ROS) are generated and released in large numbers,

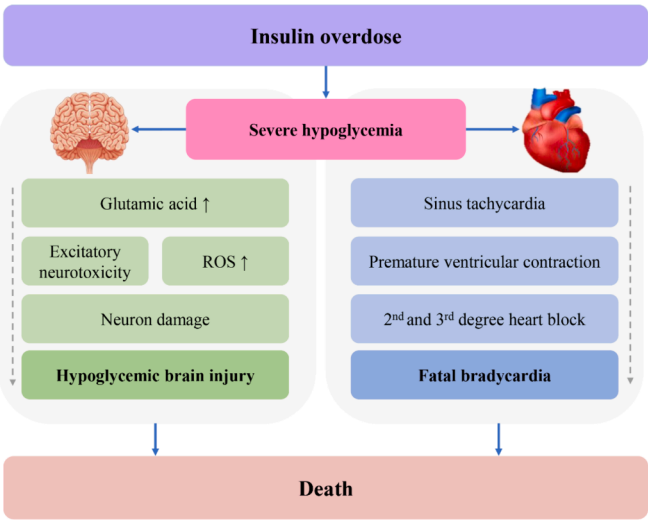


Fig. 1. Summary of potential mechanisms of death by insulin poisoning.

damaging DNA, destroying the structure and function of neurons, and eventually leading to the necrosis of neurons [24,25]. Furthermore, a transcriptome sequencing study on hippocampal transcriptome of rats with insulin-induced hypoglycemic brain injury found that differentially expressed genes were mainly concentrated in *PI3K-Akt*, *Cell adhesion molecules*, *Protein digestion and absorption* signal pathways. The transcriptional levels of *PI3K-Akt* signaling pathway and oxidative stress-related differential proteins were verified, further elucidating the important role of oxidative stress injury in hypoglycemic brain injury [15].

2.2. Cardiac death

Since the early study of Heller et al [26–28] found that hypoglycemia can lead to prolonged QT interval in the electrocardiogram, there has been increasing evidence that hypoglycemia may lead to arrhythmia and sudden death, especially in type 1 diabetes patients, whose sudden death at night accounts for 27 % of all unexplained deaths in type 1 diabetes patients [29]. The sudden death rate is 10 times higher than that of the control group of the same age [30]. This unexplained

Table 1
Information about the four common types of insulin.

Type of action	Name	Amino acid sequence	Formula	Molecular weight (Da)	Onset of action (h)	Peak action (h)	Effective duration (h)
Rapid	Aspart	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTDKT	C ₂₅₆ H ₃₈₁ N ₆₅ O ₇₉ S ₆	5826	0.15 ~ 0.25	1 ~ 2	4 ~ 6
	Lispro	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTKPT	C ₂₅₇ H ₃₈₃ N ₆₅ O ₇₇ S ₆	5807			
	Glulisine	A:GIVEQCCTSI C SLYQLENYCN B:FVKQHLCGSHLVEALYLVCGERGFF F YTPET	C ₂₅₈ H ₃₈₄ N ₆₄ O ₇₈ S ₆	5823			
Short	Recombinant Human (R)	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTPKT	C ₂₅₇ H ₃₈₃ N ₆₅ O ₇₇ S ₆	5807	0.3 ~ 0.5	2 ~ 4	5 ~ 8
	Bovine	A:GIVEQ C CA S VC S LYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTPKA	C ₂₅₄ H ₃₇₇ N ₆₅ O ₇₅ S ₆	5733			
	Porcine	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTPKA	C ₂₅₆ H ₃₈₁ N ₆₅ O ₇₆ S ₆	5777			
Medium	Recombinant Human (N)	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTPKT	C ₂₅₇ H ₃₈₃ N ₆₅ O ₇₇ S ₆	5807	2.5 ~ 3	5 ~ 7	13 ~ 16
Long	Glargine	A:GIVEQCCTSI C SLYQLENYCG B: FVNQHLCGSHLVEALYLVCGERGFF F YTPK TRR	C ₂₆₇ H ₄₀₄ N ₇₂ O ₇₈ S ₆	6063	3 ~ 4	\	24 ~ 30
	Detemir	A:GIVEQCCTSI C SLYQLENYCN B:FVNQHLCGSHLVEALYLVCGERGFF F YTPK- Myr	C ₂₆₇ H ₄₀₂ N ₆₄ O ₇₆ S ₆	5917			

Note: Differences in amino acid sequence from human insulin are marked in **bold**. “\” represents the peak of no action of long-acting insulin.

nighttime death in diabetic patients is also known as “death in bed syndrome” [31–33]. The specific mechanism of such death is not clear, but the existing research results suggest that hypoglycemia as a complication of diabetes can lead to arrhythmia, ventricular premature beat, tachycardia and heart block. Candace et al [13] constructed insulin-induced severe hypoglycemia rat model by injecting insulin into diabetic and normal blood glucose rats, and administered α/β blocker agents to analyze the symptoms of rats in each group, and found that sinus tachycardia occurred in all the dead rats, and the mortality rate of rats with tachycardia for more than 30 min was 100 %. The combination of beta blockers and alpha/beta blockers can completely eliminate hypoglycemia-induced tachycardia and mortality. In addition, the study found that in dead rats, sinus tachycardia was followed by ventricular tachycardia, with increased premature ventricular contractions and increased frequency of second degree heart blocks, further progression to third degree heart blocks associated with bradycardia, followed by reduced cardiac output, respiratory depression, hypoxemia, hypercapnia, and acidosis. Subsequently, the same research team further studied the mechanism of death of insulin-induced hypoglycemia and found that hypoglycemia could not only lead to fatal arrhythmia, but also lead to prolonged seizures mediated by the central nervous system leading to respiratory arrest, which may also be one of the pathways of hypoglycemia-related sudden death [34]. The view of death from arrhythmia caused by insulin induced severe hypoglycemia has also been reflected in other research results [10,11,35,36].

3. Forensic identification of insulin poisoning

Because there is no specific lesion of insulin poisoning and it is interfered by therapeutic insulin use, the identification of insulin poisoning cases is still difficult and challenging. According to existing studies, based on detailed case investigation, the combination of multi-organ pathological changes and in-vivo insulin test analysis plays an important role in the diagnosis of insulin poisoning. The main contents of forensic examination are shown in Fig. 2.

3.1. Case investigation

Case investigation is of great significance in the diagnosis of cause of death. Among the cases of insulin overdose deaths, suicide and homicide are more common, while accidental poisoning cases are less common [6,37]. According to a retrospective study by Stephenson et al [38], among the 40 cases of insulin overdose deaths in South Australia in the past 20 years, there were 29 suicide cases, 11 deaths were due to accidents or other causes, and 26 cases had insulin syringes or insulin injections after use at the scene. In the criminal cases of insulin homicide, Marks et al [6] found that 66 % of the murderers were medical workers or had a medical background, and 78 % of the murderers had a close relationship with the victims (such as husband, wife or lovers, etc.). In addition, in some cases, the killer will first incapacitate the victim with antipsychotic drugs, and then inject large doses of insulin. Of the seven insulin overdose deaths reported by the team over the past decade, six were homicides in which the killer had a close relationship with the victim, and in four of those cases a combination of sleeping pills and insulin was used. Therefore, in cases of suspected insulin poisoning, attention should be paid to people close to the deceased, and whether they have a medical background or have a history of exposure to insulin and sedatives.

3.2. Forensic pathology identification

3.2.1. Examination of the injection site

In normal treatment, the insulin injected into the subcutaneous tissue will gradually enter the blood circulation through the capillaries and play a role in lowering blood sugar, with almost no insulin remaining at the injection site. However, after an insulin overdose kills the body, there is usually a residue of unabsorbed insulin at the injection site. Ikeda et al [39] conducted immunohistochemistry (IHC) tests with anti-insulin antibodies to a suspect injection site with obvious bleeding in the abdomen of one victim while the IHC result of insulin is assessed in the subcutaneous tissue space. In addition, chemiluminescence enzyme

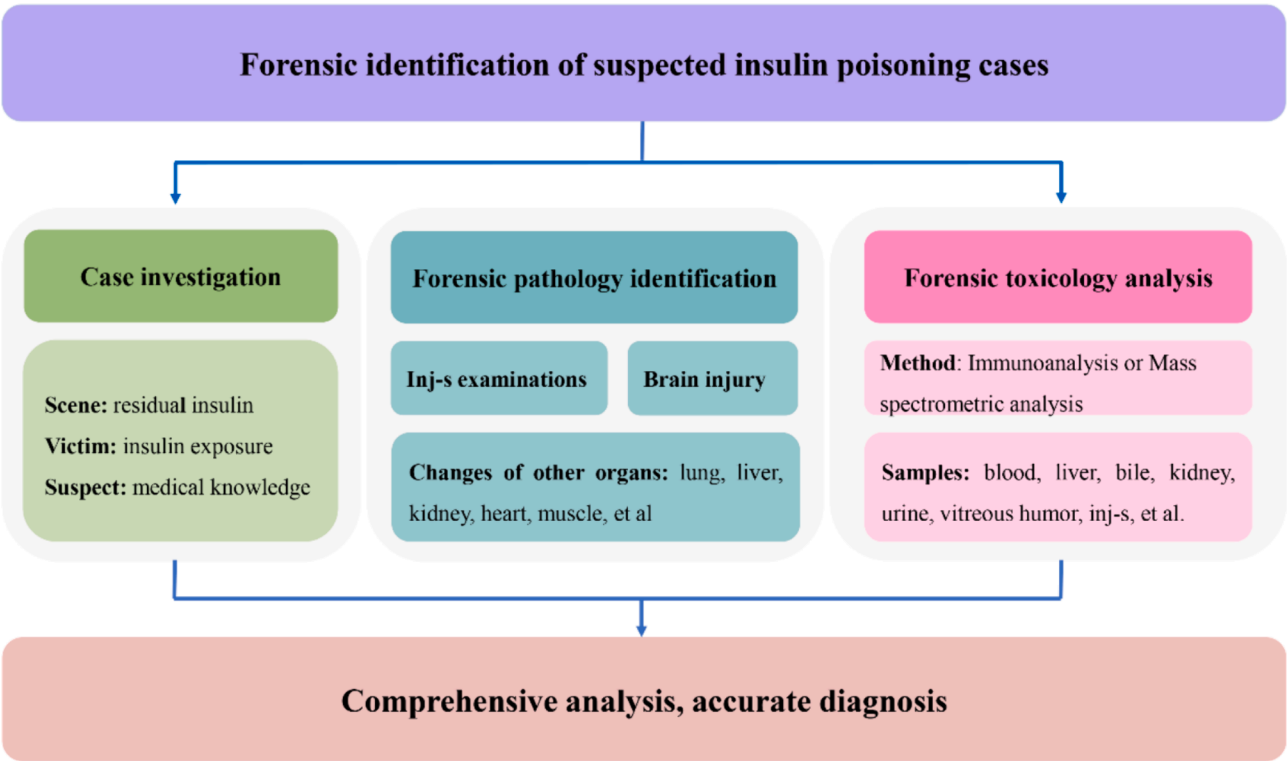


Fig. 2. Forensic identification of suspected insulin poisoning cases. (Abbreviation: inj-s: injection site).

immunoassay (CLEIA) was used to detect the insulin content at the suspected injection site in the abdomen as high as 10,000 $\mu\text{IU}/\text{mg}$, far exceeding the normal insulin secretion or therapeutic insulin injection amount in the body, which suggested an overdose of exogenous insulin. Besides, Tong et al [40] performed immunofluorescence detection of insulin and protamine in the subcutaneous tissue of 2 cases of insulin injection site by using immunofluorescence (IF) double labeling method, and the results showed that insulin and protamine were mainly distributed in the adipocyte space, and the positive detection sites of both were basically identical, which had a good detection efficiency for some insulin containing protamine.

3.2.2. Brain histopathological changes

Insulin overdose can lead to hypoglycemic brain injury, mainly manifested as brain edema. According to different sources and mechanisms, cerebral edema can be divided into vasogenic edema caused by the destruction of the blood–brain barrier and cytotoxic edema caused by brain parenchymal lesions [41]. In the study of hypoglycemic brain injury induced by insulin overdose, Tong et al [42] used diffusion weighted imaging (DWI) technology to detect multiple brain regions in rats with acute hypoglycemia, and found that apparent diffusion coefficient (ADC) in DG and hypothalamus increased, while ADC values in somatosensory layer and striatum of parietal cortex decreased. In hypoglycemic rats 7 days after glucose recovery, ADC values of DG and hypothalamus remained elevated, and ADC values of most other brain regions returned to normal, indicating that insulin-induced acute hypoglycemic brain edema was angiogenic and cytotoxic edema coexisting. Cytotoxic edema was relieved after glucose recovery, but angiogenic edema remained for a long time. This is consistent with other research conclusions [43]. In terms of cell morphological changes, the Virchow-Robin space in the cerebral cortex of rats in the acute hypoglycemia group was widened, most neurons were swollen, cytoplasm was pale, and some axons were swollen, thickened and even wavy. Hypothalamic neurons showed degeneration, deep cytoplasmic staining, and nuclear shrinkage. The neuronal swelling in CA1 and CA3 regions of hippocampus was obvious, and the granular cells of DG were dense. Severe cerebral edema, degeneration and even necrosis of neurons in DG, CA1 and CA3 regions were observed in human tissue samples from insulin overdose poisoning, and the degree of lesion was related to the duration of severe hypoglycemia. In addition, severe hypoglycemia can induce a widespread astrocyte response in the brain [40]. Although the above lesions are not characteristic, combined with the insulin residue detected at the injection site, they also contribute to the diagnosis of insulin overdose death.

3.2.3. Histopathological changes of other organs

A study [40] reported the pathological changes of multiple organs in 4 cases of insulin overdose death, and found that the main manifestations of lung tissue were edema, congestion and alveolar hemorrhage; hepatocyte edema and necrosis; the renal tubular epithelial cells were necrotic, the nucleus was dense, and part of the epithelial cells fell off into the lumen. In the heart, there were no other obvious lesions except mild inflammatory cell infiltration and hypoxic changes in cardiomyocytes. Bian et al [44] performed morphological (HE staining) and glycogen detection (PAS staining) on the liver and skeletal muscle of rats with insulin poisoning death and recovery, and the results showed that the central vein and hepatic sinus of rats with insulin poisoning death were obviously congested, and the glycogen content in liver and skeletal muscle was significantly reduced. However, in the recovery group of insulin poisoning, hepatocyte swelling and ballooning were obvious, and the glycogen content in liver and skeletal muscle was significantly increased, which could lead to the occurrence of glycogenic liver disease. In addition, it has been reported that insulin overdose can lead to heart block [11,34] but its diagnosis is usually dependent on clinical examination, and no obvious pathological changes have been found, which requires further study. Although the above pathological changes

have some manifestations in the body of insulin poisoning, the lack of specificity lead to that they will not be used as diagnostic markers in the identification of related cases, but can play an auxiliary diagnostic role.

3.3. Forensic toxicology analysis

3.3.1. Detection methods

Fasting plasma insulin concentration in healthy adults is about 5 ~ 15 $\mu\text{U}/\text{mL}$ (30 ~ 90 pmol/L), and the concentration increases 4 ~ 5 times after diet [1]. Elevated fasting insulin levels usually indicate the presence of islet cell tumor, insulin resistance due to type 2 diabetes, or other chronic metabolic diseases, while decreased levels are usually seen in type 1 diabetes. At present, there are various methods for insulin concentration detection. According to different detection principles, it can usually be divided into immunoassay and mass spectrometry:

3.3.1.1. Immunoassay analysis. Radioimmunoassay (RIA), enzyme-linked immunosorbent assay (ELISA) and chemiluminescence immunoassay (CLIA) are the earliest and most widely used methods for the detection of insulin in clinical and forensic practice. These three detection methods are based on the basic principle of antigen antibody binding, through different detection techniques to complete the determination of insulin concentration in vivo. As early as the last century, researchers have used these three technologies to develop human insulin detection kits, which can detect insulin in human serum or plasma with a sensitivity of 1 pmol/L , far lower than the fasting insulin secretion level of healthy adults [45–48]. However, due to the limitation of the detection principle, such kits may also detect other insulin analogues or proinsulin as insulin precursors that are basically consistent with human insulin sequence, which may lead to errors in the detection results [49]. In forensic identification, although such kits have the ability to quantitatively detect exogenous insulin, their specificity is poor, it is difficult to distinguish other types of insulin analogues, and there are certain evidence defects in case detection. In addition, the immunological method has a great impact on the quality of the tested material, and if the blood sample has hyperlipidemia, jaundice, excessive protein components or hemolysis, the test results may not be reliable enough. At the same time, most of the kits did not perform method validation for insulin detection in other samples (such as vitreous fluid, bile, tissue homogenate), so caution should be exercised when interpreting the results of insulin measurement in such samples, which has limitations in forensic practice.

3.3.1.2. Mass spectrometry. Mass spectrometry is an analytical method to measure the mass charge ratio of ions. Its basic principle is that each component in the sample is ionized by the ion source to generate charged ions with different mass charge ratio, and the ion beam is formed under the action of accelerating electric field, which is detected and distinguished by the mass analyzer and detector. In addition, chromatographic technology can separate different components in complex mixtures, combined with the application of chromatography and mass spectrometry, can easily separate and qualitative and quantitative detection of different components in the sample, in the field of biological macromolecule analysis has gradually been widely used. Some methods for insulin detection based on liquid chromatography-tandem mass spectrometry (LC-MS/MS) have been reported. Bottinelli et al [50] used liquid chromatography-high resolution mass spectrometry (LC-MS/HRMS) to spectrometry platforms, seven kinds of insulin detection methods were established in human plasma (living body) and serum (postmortem), including Human Insulin and Insulin Lispro, etc. The detection limits were up to 0.5 ~ 10 ng/mL . Subsequently, the team further optimized the method and detected a variety of insulins in other organs and tissues of the human body (such as liver, kidney, muscle and subcutaneous tissue at the insulin injection site), with the detection limit up to 25 ng/mL [51], providing a new detection method for the forensic diagnosis of death from insulin injection overdose. However, despite the

high specificity and sensitivity of insulin detection based on mass spectrometry technology, the pre-treatment methods of different tissues are demanding, the operation process is complex, and the detection cost is high. In addition, at present, there is still a lack of corresponding diagnostic standards and norms in the world, and there are still some limitations in the practical application of forensic medicine. Table 2 summarizes the research on insulin detection methods in various samples based on LC-MS/MS technology for reference in insulin detection by mass spectrometry.

3.3.2. Selection of test samples

Because of the different ingestion routes, body distribution and main metabolic ways of different types of poisons (drugs), it is very important to choose the right type of toxicant analysis samples. In terms of the selection of materials for the case of insulin overdose death, there are mainly the following:

3.3.2.1. *Blood.* Blood is one of the most common forensic toxicology testing samples. Various detection techniques have been used for the qualitative and quantitative analysis of insulin in postmortem blood. For example, from the RIA technology used by Iwase et al [74] in the early stage to the chemiluminescence enzyme immunoassay (CLEIA) by Palmiere et al [75] and the LC-MS/HRMS by Bottinelli et al [50], all of them provide an important means for the quantitative analysis of insulin in blood after death. However, in some cases of coagulation, hemolysis and spoilage, especially in cases of hemolysis, the insulin content in blood is usually not accurately measured due to the degradation of insulin by hemoglobin after hemolysis [76], and there is a big difference between it and the actual concentration. At this time, other test materials can be considered for analysis.

3.3.2.2. *Liver.* After endogenous human insulin is secreted from the pancreas, it is first transported to the liver through the hepatic portal vein for the first pass metabolism, and about 50 % of endogenous human insulin is eliminated in this process, so the liver is considered to be the main organ of insulin metabolic elimination [77]. At present, there are few studies on quantitative detection of insulin in the liver. Bottinelli et al [51] used LC-MS/HRMS technology to detect insulin in liver tissue homogenates after immunopurification, and successfully detected winter insulin for the first time in 2 cases of liver tissues killed by insulin overdose, with a detection limit of 25 ng/g. However, the method failed to detect endogenous insulin in 5 cases of liver tissue that did not receive insulin injection, possibly because the detection limit was much higher

than the insulin concentration in normal human liver tissue. In addition, Dickson [78] and Hess et al [60] used RIA and LC-MS/MS methods to detect insulin in liver, respectively, but the results were negative, reflecting that liver has certain limitations in insulin detection.

3.3.2.3. *Bile.* As early as 1972, Sturner et al [79] used RIA technology to detect the insulin concentration in postmortem bile of 21 cases (20 of which had no insulin exposure history and 1 case died 9 days after injecting human insulin), and found that human insulin was successfully detected in 16 of the 20 bile samples without insulin exposure history. Its concentration ranges from 6.5 to 487.5 $\mu\text{U/mL}$ ($1\text{ U}\approx 0.035\text{ mg}$), and it can be detected even in the bile of partially decomposed cadavers. However, in the bile of a patient who died 9 days after injecting human insulin, the concentration of human insulin was detected as high as 768 $\mu\text{U/mL}$, which was much higher than the concentration of human insulin in the serum of this case of 42.5 $\mu\text{U/mL}$ [77]. In addition, Palmiere et al [75] reported that in a case of death 18 h after insulin injection, the insulin concentration detected by CLEIA method was 19.80 $\mu\text{U/mL}$ in bile, which was lower than the serum insulin concentration of 582.50 $\mu\text{U/mL}$. Meanwhile, the same study [75] also reported that the insulin concentration in the bile of 20 patients with insulin-dependent diabetes mellitus and 20 patients without diabetes history within 12 to 20 h of death was lower than that in the corresponding serum (insulin dependent diabetes mellitus: bile/serum = 1.4/26 $\mu\text{U/mL}$; No diabetes: bile/serum = 1.1/2.4 $\mu\text{U/mL}$), suggesting that there may be a certain temporal change in the ratio of insulin concentration in serum and bile, and insulin concentration in bile seems to be less affected by insulin-dependent diabetes, but there are no systematic studies, which can be further studied. At present, there are no reports on the detection of insulin in bile by mass spectrometry.

3.3.2.4. *Kidney.* Unlike the metabolic pathway of endogenous human insulin, exogenous insulin is absorbed into the blood circulation after subcutaneous injection without first passing elimination by the liver [4]. According to existing studies, compared with liver, the possibility of detecting insulin in kidney is greater than that in liver due to the presence of glomerular nonspecific diffusion, specific receptor-mediated transport filtration effect and proximal renal tubule reabsorption [77]. Bottinelli et al [51] used LC-MS/HRMS techniques to successfully detect insulin positivity in 5 out of 13 postmortem kidney cases involving insulin, and only 1 of them also detected insulin in the liver. Hess et al [60] also detected insulin by LC-MS/MS technique in the kidney of a suicide patient who over-injected insulin, with a concentration of 384 $\mu\text{U/g}$, but not in the liver. Insulin concentrations in different areas of the kidney also seem to vary. Thompson et al [80] used in vitro autoradiography to detect ^{125}I labeled insulin in different regions of the kidney, and found that the distribution of insulin in the kidney cortex was more dense and extensive than that in the medulla. The analysis suggested that the distribution might be related to the fewer insulin receptor binding sites in the medulla of the kidney, which provided an important reference for selecting appropriate test material areas in the actual case.

3.3.2.5. *Urine.* Because insulin has a short half-life in the blood and is reabsorbed by the renal tubules, there is rarely insulin in complete form in the urine, which makes it very difficult to quantitatively analyze it [6]. In the forensic identification of fatal cases of insulin overdose, there are few reports on the detection of insulin in urine. However, in the field of doping control, detecting insulin in urine is of great significance. In a study in 2006, Thevis et al [81] successfully detected fast-acting insulin analogues in urine after solid phase extraction combined with immunopurification through LC-MS/MS technology, and the detection limit reached 50 pg/mL. Subsequently, the team further developed six fast-acting, medium-acting and long-acting insulin detection methods for urine, with detection limits up to 10 ~ 25 pg/mL [61]. Judak et al [66] raised the detection limit of insulin in urine to 10 pg/mL by using LC-

Table 2
Summary of insulin detection by LC-MS/MS.

Samples	Pretreatment method	LOD	References*	Other researches#
Blood	PPT + FTT + IP	0.5 ng/mL	[50]	[52–59]
Liver	IP	25 ng/g	[51]	\
Kidney	IP	25 ng/g	[51]	[60]
Muscle	IP	25 ng/g	[51]	[60]
Urine	SPE	0.01 ng/mL	[61]	[62–69]
Vitreous humor	PPT + SPE	0.1 ng/mL	[70]	[71,72]
Blood spot	SPE + IP	0.5 ng/mL	[73]	\
Inj-S	PPT + IP	20 ng/g	[60]	[51]

Note: (Abbreviation: LOD: limit of detection; Inj-S: injection site; PPT: protein precipitation; FTT: filtration; IP: immunopurification; SPE: solid phase extraction;) (“*” indicates the source literature of the pre-treatment method and detection limit of each test material; “#” indicates other studies that used this type of test material, and the pre-treatment method and detection limit may be different from the contents shown in the table; “\” indicates that no such document is available.).

MS/HRMS technology with higher resolution and sample pre-treatment method based on ultrafiltration combined with immunopurification, which was consistent with other research results. In order to solve the problem that the complete insulin content in urine is low and difficult to detect, Thevis [64,65] research team analyzed insulin metabolites in urine. Human insulin metabolites DESB30, DESB24-30 and DESB25-30 as well as metabolites DESB24-32, DESB30-32 and DESB31-32 of insulin glargine were successfully detected, providing a new method and idea for the detection of insulin and metabolites in urine. However, this method requires a large urine sample size, usually more than 20 mL, and may be difficult to collect adequately in some cases of death.

3.3.2.6. Vitreous humor. Due to the isolation effect of blood-eye barrier, substances in vitreous body fluids are less affected by hemolysis and have high stability. In terms of the detection of insulin in vitreous body fluids, some scholars have carried out research through a variety of techniques. Skowronek et al [82] used RIA method to determine insulin concentration in 93 cases of vitreous body fluids, the detection limit was 0.5 $\mu\text{U/mL}$, and 7 cases were positive, with the concentration range of 1.42 ~ 24.42 $\mu\text{U/mL}$. In the vitreous body fluids of 10 patients with insulin-dependent diabetes mellitus and 10 patients with no history of insulin exposure, the average insulin concentration measured by Palmieri et al [75] using CLEIA method was 2.20 $\mu\text{U/mL}$ and 0.2 $\mu\text{U/mL}$, respectively. Significantly lower than the mean serum insulin concentrations of 26 $\mu\text{U/mL}$ and 2.4 $\mu\text{U/mL}$. The concentrations of insulin in serum and vitreous body fluids were 582.5 $\mu\text{U/mL}$ and 11.5 $\mu\text{U/mL}$, respectively, in a patient with insulin poisoning. However, Bugelli et al [83] even used CLEIA method to detect higher insulin concentrations of 74.23 $\mu\text{U/mL}$ (left) and 61.11 $\mu\text{U/mL}$ (right) in bilateral vitreous body fluids of a sublingual suicide patient, which is of great significance for the diagnosis of insulin poisoning. Beckett et al [70] used LC-MS/HRMS to successfully detect winter insulin, insulin glargine and its metabolite M1 in vitreous fluids of 3 cases of insulin poisoning, with a detection limit of 0.1 ng/mL. The above studies provide a variety of methods for the analysis of insulin and its metabolites in vitreous body fluids. However, it should be noted that the insulin content in vitreous body fluids is much lower than that in serum regardless of whether the death is insulin-related. Control experiments should be fully carried out in actual detection to prevent misjudgment of results.

3.3.2.7. Inj-S. In a fatal overdose of exogenous insulin, blood circulation stops and there is usually a large amount of unabsorbed insulin left at the injection site. Tong et al [40] successfully detected insulin and protamine in subcutaneous tissues at the injection site by using immunohistochemistry and immunofluorescence double labeling techniques, which visually showed insulin deposition at the injection site. However, this method could not quantitatively analyze insulin concentration, which had certain limitations. Ikeda et al [39] used CLEIA method to detect the insulin content at the injection site of the deceased with insulin poisoning, and found that the insulin concentration in the subcutaneous adipose tissue at the injection site was as high as 10,000 $\mu\text{U/mg}$, which was much higher than the insulin concentration in other samples such as blood, urine and bile in the same case. By using LC-MS/MS, Hess et al [60] detected that the insulin concentration in the subcutaneous tissue of a patient with insulin poisoning at the injection site was 808 $\mu\text{U/g}$, which was also much higher than the insulin concentration in other samples. Although the concentration of insulin detected at the injection site is usually much higher than that of other samples, it is also affected by the false-positive results caused by therapeutic insulin injection. In actual cases, it is usually necessary to combine the results of other types of samples to synthesize the case data and make accurate judgments.

4. Conclusion

In this paper, the poisoning mechanism of insulin overdose death, the change of forensic pathology, insulin detection technology and the research status of insulin detection in various forensic materials were reviewed and analyzed. Excitatory neurotoxicity and hypoglycemic brain injury due to oxidative stress and arrhythmia due to heart block are generally considered to be the main lethal mechanisms of insulin overdose, which can lead to death either alone or in combination. In the forensic medical identification of suspected cases of insulin overdose death, detailed case investigation should first be carried out to determine the research direction; secondly, pathological examination of multi-organ tissues can directly or indirectly determine whether there is exogenous insulin intake; finally, immunoassay or mass spectrometry can be used to directly qualitatively and quantitatively analyze insulin concentration in various samples. Determine whether there is exogenous insulin overdose. Appropriate analysis methods and types of materials can be selected according to the circumstances of the case in order to better serve the forensic medical identification of insulin overdose death cases.

Ethics approval

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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